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LIST OF SYMBOLS AND ABBREVIATIONS

SYMBOLS	ABBREVIATION
A/V	Artery/Vein
AAA	Artificial Algae Algorithm
AAASVM	Artificial Algae Algorithm with Support Vector Machine
AI	Artificial Intelligence
AMD	Age-Related Macular Degeneration
ANN	Artificial Neural Network
AUC	Area Under Curve
BOSVM	Bayesian Optimization Support Vector Machine
CAD	Computer Aided Design
CAV	Convolution Along Vessel
CDR	Cup Disc Ratio
CNN	Convolutional Neural Networks
CNN-LSTM	CNN-Long Short-Term Memory
CoG-NET	Classification of Glaucoma Network
CWS	Cotton Wool Spots
CycleGAN	Cycle Generative Adversarial Network
DCNN	Deep Convolutional Neural Network
DED	Diabetic Eye Disease
DeiT	Data-Efficient Image Transformer
DFP	Digital Fundus Photography
DL	Deep Learning
DM	Diabetic Mellitus
DME	Diabetic Macular Edema
DNCN	Deep Neural Classification Network
DNN	Deep Neural Network
DOMLS	Deep Optimized Machine Learning Strategy
DPA	Depthwise Parallel Attention

DR	Diabetic Retinopathy or Diabetic Retinal Disease
DRIVE	Digital Retinal Image Vessel Extraction
DS	Decision Stumps
DT	Decision Tree
EAIHSEC	Explainable Artificial Intelligence Hybrid Stacking Ensemble Classification
ECNet	Evolutionary Convolutional Network
EX	Exudates
FANet	Fully Attention based Network
FJAM	Fuzzy Joint Attention Module
FN	False Negative
FP	False Positive
FRCNN	Faster Region Based Convolutional Neural Network
FRCNNAAASVM	Faster Region-based Convolutional Neural Network- Artificial Algae Algorithm Support Vector Machine
GLCOM	Grey Level Co-occurrence Matrix
GLRLM	Grey Level Run Length Matrix
GON	Glaucomatous Optic Nerve
HEM	Hemorrhage
HRF	High Resolution Fundus
HSEC	Hybrid Stacking Ensemble Classifier
IoP	Intraocular Pressure
LBP	Local Binary Patterns
LQ	Low Quality
MA	Microaneurysms
MCG-Net	Multilabel Convolutional Graph Network
MKMOL	Modified K-Means Optimization Logic
ML	Machine Learning
MLP	Multi Layer Perceptron
MSE	Mean Square Error
NIO	Neural Image Optimization

NN	Neural Network
OAL	Ocular Axial Length
OC	Optical Cup
OCT	Optical Coherence Tomography
OD	Optical Disc
OHTS	Ocular Hypertension Treatment Study
ONH	Optical Nerve Head
OWDANN	Optimal Weighted Based Deep Artificial Neural Network
POAG	Primary Open Angle Glaucoma
RBF	Radial Basis Function
RDR	Rim Disc Ratio
ReLU	Rectified Linear Unit
RF	Random Forest
RMSE	Root Mean Square Error
RNFL	Retinal Nerve Fiber Layer
ROC	Receiver Operating Characteristic
ROI	Region of Interest
RSOAE	Random Search Optimization Adaboost Ensemble
SAP	Standard Automated Perimetry
SLO	Scanning Laser Ophthalmoscopy
SMO	Sequential Minimal Optimization
SOA	Swarm Optimization Algorithm
SURF	Speeded Up Robust Features
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
VCDR	Vertical Cup Disc Ratio
ViT	Vision Transformer
XAI	Explainable Artificial Intelligence

SYMBOLS

i	Hyperparameter
$f(i)$	Function
Θ	Parameter Space
x_a, y_a	Variable Ranges
s	Samples
$g_z(i)$	Weak Classifier
ε_z	Weighted Classification Error
x, y	Image Intensity
H	Feature Map
k	Layer
C	Convolution Kernal
b	Bias
σ	Activation Function
$\mathcal{L}$	Loss Function
η	Learning Rate
$\hat{S}$	Predicted Segment S
S	Ground Truth
$\nabla_{\theta} \mathcal{L}$	Loss Gradient
W	Weight
β	Hyper Parameter
$x_i x_j$	Kernal Input Samples
γ	Positive Scalar
Y_i	Algae Population
NP	Total Number of Algae in Population
D	Size of Algae colony
r_j	Number of Classifiers
D	Distance
P	Cooccurence Matrix
$\Delta x, \Delta y$	Offset of Neighbour Pixel

gl	Grey Level
rl	Run Length
Mg	Maximum Gray
$Smax$	Maximum Length
$\bar{A}$	Standard Mean
h_i	Dataset
X'_i	Feature Vector
$H(x)$	Hybrid Classifier